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# Preparation of geopolymer insulating bricks from waste raw materials

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## HIGHLIGHTS

- Insulating geopolymer samples were prepared using industrial wastes raw materials.
- Aluminum reacted with sodium and calcium hydroxides to produce hydrogen.
- Pores were formed in the geopolymer matrix owing to the evolution of hydrogen.
- Porosity reached 50% on using 5% aluminum trimmings and 15% dealuminated kaolin.
- Compressive strength of bricks was 1.4 MPa and thermal conductivity 0.26 $W.m^{-1}.K^{-1}$.

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## ABSTRACT

Insulating geopolymer samples were prepared from cheap sources, namely, the powder residue from the production of clay bricks (clay bricks waste), slaked lime waste from acetylene production and waste aluminum trimmings from workshops. 0.5% caustic soda was used, together with slaked lime as alkaline activator. De-aluminated kaolin, a waste from the alum industry, was also added as binding agent to substitute part of the clay bricks waste. Hydrogen produced from reaction between aluminum and alkalis induced the formation of pores inside the geopolymer matrix. It was found that using 5% (by weight) aluminum trimmings and substituting 15% of clay bricks waste by de-aluminated kaolin was enough to raise the porosity above 50%, thus producing light bricks of densities in the 1000 $kg.m^{-3}$ range. The compressive strength of these bricks was about 1.4 MPa and their thermal conductivity as low as 0.26 $W.m^{-1}.K^{-1}$.


## 1. Introduction

Since the discovery of the geopolymerization reaction by Davidovits [1], this novel material has come to be a serious competitor for conventional cement based civil works. Since then, it has been used in the production of different types of bricks [2–6], roofing tiles [7], concretes [8–10], road paving material [11,12] and even in aircraft pavements [13]. The preparation of geopolymer bodies relies essentially on reacting an alkaline activator with a binding material. The former material usually consists either of metakaolin, obtained by the calcination of kaolin above 600 °C, or fly ash. The binder category also included iron slag [14,15], silica fume [16,17] and red mud [18,19]. Recently, other binding materials have been used such as clay bricks waste [20], or ceramic tiles dust waste [21]. Alkaline activators, on the other hand, usually consist of caustic soda and sodium silicate in different ratios so as to preserve a certain total sodium oxide content. In this respect, Zhuang et al. [22] as well as Degirmenci [23] outlined that an increase in silicate to NaOH ratio was associated with increased strength of the resulting geopolymer. Other activators such as slaked lime are usually used in connection with caustic soda which acts as main activator [24,25] or as mixed with Portland cement [26].

The scope of use of geopolymeric materials has seen much novel applications such as fire resistant geopolymer [27,28], acid resistant mortars and concretes [29–31] as well as insulating bricks. Many studies were carried out to investigate the preparation geopolymers for use as insulating components in households [32–40]. They relied on using foaming agents as pore generating materials. Ibrahim et al. [32] used an unidentified foaming agent in proportions from 5 to 10% to prepare fly ash based geopolymer bricks using caustic soda and sodium silicate as activating solution. Their samples displayed densities in the 1400–1500 $kg.m^{-3}$ range associated with compressive strengths varying from 5 to 10 MPa depending on the percentage of foaming agent added, although these are too high densities for insulating components. Similar

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results were also obtained by Risdanareni et al. [33] using Styrofoam as pore generating agent in levels up to 0.9%. On the other hand, Roviello et al. [34] were able to prepare hybrid geopolymer-based foams with lower densities (250–850 kg/m$^3$) that showed good mechanical properties, fire resistance and low thermal conductivity. Also, hydrogen peroxide was used as foaming agent to generate porosity during the preparation of geopolymers [37–40]. In general, however, the elevated molarity of NaOH used (Usually >10 M) poses a serious economic constraint on the application of these technologies. This urged the need for using low–cost materials to produce geopolymer foams such as waste glass [38,41], polyurethane waste [42] and red mud [41].

In the present paper, is suggested an economic method of producing low density insulating geopolymer bricks using waste slaked lime together with a minimal amount of caustic soda as activator and two solid pozzolanic wastes, namely, fines produced from clay brick manufacture (clay bricks waste) and de-aluminated kaolin waste from alum production as binders. As for porosity, it is generated through the action of both slaked lime and sodium hydroxide on aluminum trimmings wastes.

## 2. Experimental

### 2.1. Raw materials

Five raw materials were used in this work:

1. Ground waste produced from firing clay bricks (clay bricks waste) that is used as a substitute for the much more expensive calcined kaolin. The waste was sieved over 200 meh screen (74 μm) and the undersize used in the preparation of the samples.
2. Commercial calcium hydroxide waste resulting from the manufacture of acetylene. It was received as 8% milk lime suspension.

   This waste is produced as by product from acetylene preparation using calcium carbide:

   $CaC_2 + 2\ H_2O = C_2H_2 + Ca(OH)_2$

   According to the Egyptian Public Authority for Industry, the amount of acetylene produced locally in 2017 was 360 tons corresponding to about 1000 tons of slaked lime. This latter is usually commercialized at low prices in the building industry.

3. Commercial sodium hydroxide powder was supplied by Morgan Company, 10th of Ramadan City, a Cairo suburb.
4. Aluminum waste trimmings resulted from workshops for aluminum products' manufacturing. These are scattered all over the country and a typical workshop would yield about 10–20 kg per day. These are simply swiped out as garbage. These were screened to pass 20 mesh screen (0.833 mm).
5. Alum waste supplied from The Aluminum Sulfate Company of Egypt, located in a Cairo suburb. Its screen analysis is shown in Fig. 1. Its median size ($D_{50}$) is 0.21 mm. This is produced by the action of conc. sulfuric acid (98%) on kaolin calcined at 700 °C. Alum waste consists of the solid residue left after the reaction, whereas alum (aluminum sulfate) is obtained as solution.

$$Al_2O_3.2SiO_2{\cdot}2H_2O = Al_2O_3.2SiO_2(\text{m-kaolin}) + 2H_2O \quad (1)$$

$$3H_2SO_4 + Al_2O_3.2SiO_2 = Al_2(SO_4)_3(\text{sol.}) + 3H_2O + 2SiO_2 \quad (2)$$

The annual production of alum (aluminum sulfate) in Egypt in 2017 was about 240,000 tons mainly used in water treatment.

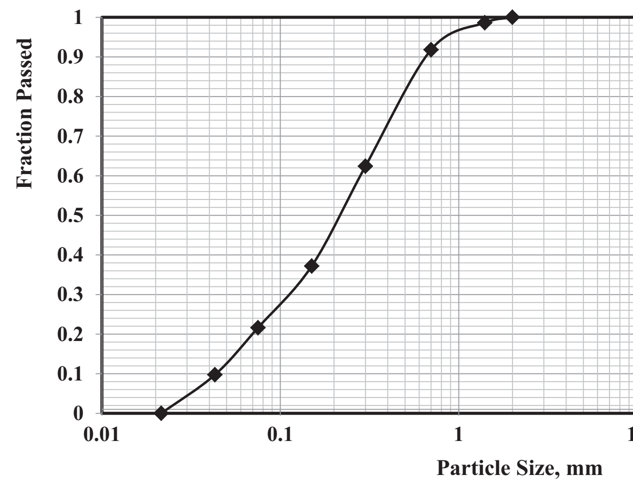



**Fig. 1.** Particle size distribution of de-aluminated kaolin.

The corresponding amount of de-aluminated kaolin waste is about 80,000 tons which are dump filled in desert areas.

The principle of pore generation is based on the reaction of both calcium hydroxide (slaked lime) and NaOH with aluminum metal [43]. The release of hydrogen is responsible for pores formation.

$$2Al + 3Ca(OH)_2 + 6H_2O = Ca_3Al_2[(OH)_4]_3 + 3H_2 \quad (3)$$

$$2Al + 2NaOH + 2H_2O = 2NaAlO_2 + 3H_2 \quad (4)$$

### 2.2. Characterization of raw materials

The phase composition of solid phases was assessed by XRD using Brukur $D_8$ Advanced Computerized X–Ray Diffractometer apparatus with mono–chromatic Cu $K_\alpha$ radiation, operated at 40 kV and 40 mA. XRD has revealed that the main phase present in clay bricks waste is quartz. This is due to the fact that clay bricks are usually fired below 900 °C so that they will be constituted mainly from quartz and amorphous meta-kaolin. Quartz was also the main phase appearing for de-aluminated kaolin. This is since this waste is the residue remaining after treating meta-kaolin with sulfuric acid to produce aluminum sulfate that is removed as solution (Eq. (2)). Unreacted meta-kaolin being amorphous does not produce any XRD lines while some undissolved aluminum sulfate produced weak diffuse peaks of the octahydrate.

As for chemical composition, it was determined by means of an AXIOS, panalytical 2005, Wavelength Dispersive (WD–XRF) Sequential Spectrometer. A JEOL–JSM 6510 apparatus with zoom magnification power up to 300,000× was used to generate SEM images for some selected specimens. Table 1 reveals the chemical analyses of the two binding materials used.

The powder density of different brick samples was determined by grinding down the brick and using the density flask method following ASTM B 311 [44].

### 2.3. Preparation of geopolymer brick samples

Following the work of Selem et al. [25], 8% milk of lime suspension was mixed with waste ground clay brick dust and water in relative percentages of 9:63:28% by weight respectively. To the mixture, caustic soda was added as 0.5% of the total mass. After mixing for 30 min, the mixture was poured in 50 × 50 × 50 mm$^3$ steel molds. The brick samples were left in the molds for 24 h. They were then taken out of the molds to cure for up to 28 days at temperature of 28 ± 2 °C and 40 ± 3% humidity.

**Table 1**
Chemical composition of clay bricks dust waste (homra) and de-aluminated kaolin.

| % By weight | $SiO_2$ | $Al_2O_3$ | $TiO_2$ | $Fe_2O_3$ | CaO | $Na_2O$ | $K_2O$ | $SO_3$ | LOI | Total |
|---|---|---|---|---|---|---|---|---|---|---|
| Homra | 65.83 | 16.55 | 0.87 | 7.97 | 3.9 | 1.59 | 1.31 | – | 1.95 | 99.97 |
| DeAl kaolin | 74.86 | 8.48 | 3.95 | 1.51 | 0.32 | 0.11 | 0.08 | 6.28 | 4.02 | 99.61 |

Ground clay brick waste was substituted by aluminum fillings for up to 15% of its weight whereas its substitution by de-aluminated kaolin ranged from 0 to 20%.

### 2.4. Characterization of prepared geopolymer samples

The bulk density was determined geometrically as the quotient of the mass of the brick by its bulk volume.

The compressive strength of the prepared geopolymer bricks was determined according to ASTM C 67 [45]. Three specimens were tested each time at uniform rate (2.5 $kN.s^{-1}$) using a compressive strength machine of maximum load capacity of 1.5 MN. The degree of geopolymerization was calculated following the method suggested by Nikolic et al. [46] as follows:

A geopolymer sample ($m_{sample}$) was dissolved in HCl. The remaining solid was separated by filtration, washed with water to remove excess HCl and then dried. The solid was treated with $Na_2CO_3$ and then another separation procedure was performed. The mass of final solid residue ($m_{residue}$) (unreacted material) was determined. The degree of geopolymerization GD can be calculated from the following equation:

$$GD = \frac{M_{sample} - [M_{residue} \times (1 - LOI)]}{M_{sample}} \times 100\% \quad (5)$$

Where:

$M_{sample}$: the mass of the powder sample (g)

$M_{residue}$: the mass of the residual solid (g)

LOI: the loss on ignition of the powder sample.

In all cases, a sample consisting of three specimens was tested and the average value recorded.

Finally, the thermal conductivity of the prepared samples was determined using a KD2 Pro Thermal Properties Analyzer that uses a transient heat conduction algorithm to digitally record thermal conductivity at room temperature.

## 3. Results and discussion

### 3.1. Effect of addition of aluminum trimmings on the properties of geopolymer samples using clay bricks dust as binder

The first experiments were performed with clay bricks dust as sole binder. Geopolymer samples prepared by adding different levels of aluminum trimmings showed a regular pattern consisting of a drop in bulk density following an increase in aluminum percentage and prolonged curing. However, it seems that the reaction between aluminum and slaked lime and caustic soda was initially very rapid as evidenced by the sharp drop in density observed after 3 days. Subsequent curing did not appreciably affect the values of bulk density (Fig. 2), proving that the reactions associated with hydrogen evolution were completed. On the other hand, Fig. 3 reveals that porosities as high as 60% could be achieved on adding 15% aluminum trimmings.

On testing the prepared samples for compressive strength after different periods of curing, it was evident that the formation of pores following the evolution of hydrogen from Eq. (1) had an extremely negative effect of strength. In any case, curing time did not seem to seriously affect the values of strength (Fig. 4). A comparison between this latter Figure and Fig. 2 shows that while the use of 15% aluminum results in bulk densities in the 850 kg.

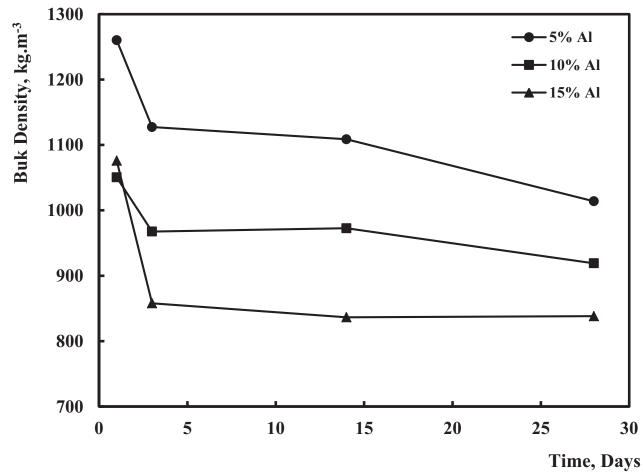



**Fig. 2.** Effect of adding aluminum trimmings on the bulk density of geopolymer samples using clay bricks dust as binder (0% DK).

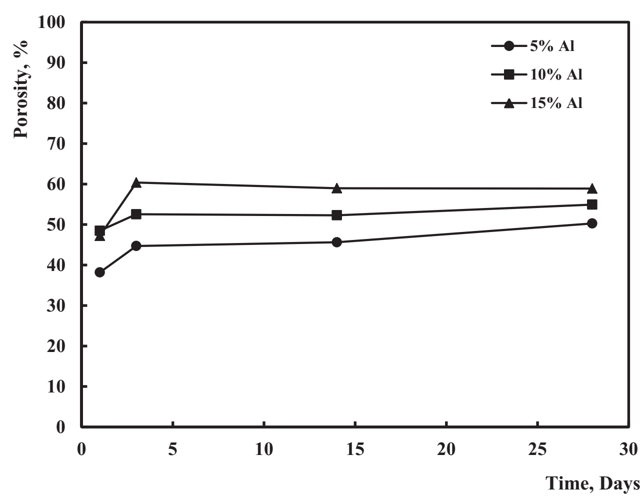



**Fig. 3.** Effect of adding aluminum trimmings on the porosity of geopolymer samples using clay bricks dust as binder (0% DK).

$m^{-3}$ range, the compressive strength drops to less than 0.4 MPa which resulted in friable samples.

This has prompted the use of a pozzolanic waste that could react in alkaline medium to produce hydrates that can add to the strength of the geopolymers. The chosen waste was de-aluminated kaolin which, as previously stated, mainly consists of silica in fine form. The reaction with calcium hydroxide in aqueous medium produces calcium silicate hydrate (CSH) which is known to have a positive impact on increasing the bricks strength [47].

### 3.2. Effect of substituting clay bricks dust by de-aluminated kaolin

As clay bricks dust was gradually substituted by de-aluminated kaolin (DK), the bulk density followed a more or less fixed pattern revealed in Fig. 5, which was drawn for a sample containing 5% aluminum trimmings. While DK starts its pozzolanic reaction with

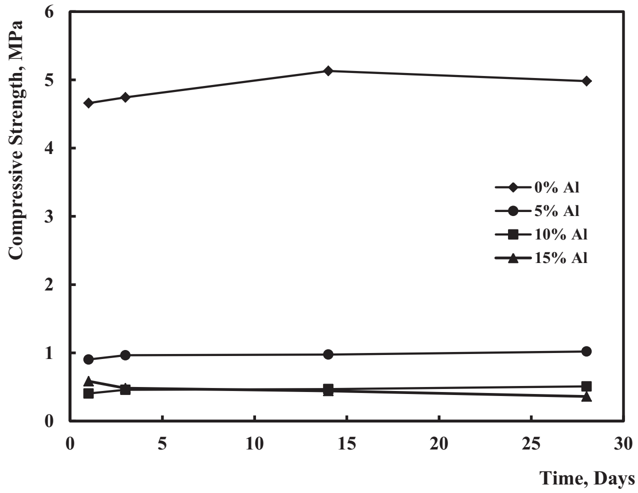



**Fig. 4.** Effect of adding aluminum trimmings on the compressive strength of geopolymer samples using clay bricks dust as sole binder.

lime, aluminum starts reacting with alkaline solutions to produce hydrogen (reactions (1) and (2)). The net effect is to cause a decrease in bulk density that becomes less effective as more DK is used owing to the formation of CSH needles which tend to close the pores. Little change was observed after 7 days curing and the final values of density ranged from 960 to 1000 kg.m$^{-3}$ except for the sample containing 20% DK where it reached 1053 kg.m$^{-3}$. In Fig. 6, the SEM micrograph illustrates the presence of CSH needles obtained when a specimen containing 5% aluminum trimmings and 15% de-aluminated kaolin was left to cure for 7 days. The figure also reveals the presence of some porosity in the matrix.

When more aluminum was added, the effect of pores formation exceeded that of DK reacting with lime so that the bulk densities steadied to more or less fixed values at about 7% addition (940–970 kg.m$^{-3}$) after an initial drop (Fig. 7). The corresponding figure on adding 15% aluminum gave similar results where the final densities were in the same previous range, indicating that exceeding 10% aluminum produced no more pores due to exhaustion of calcium hydroxide.

The effect of substituting clay bricks dust by DK was very pronounced in increasing the compressive strength of geopolymer samples. In general, an increase in porosity is usually accompanied by a decrease in strength. In a comprehensive review of the subject, Chen and Zhou [48] suggested more than one form of dependence between strength ($\sigma$) and porosity ($p$), such as:

$$\sigma = \sigma_0(1-p)^n \tag{6}$$

$$\sigma = \sigma_0 e^{-kp} \tag{7}$$

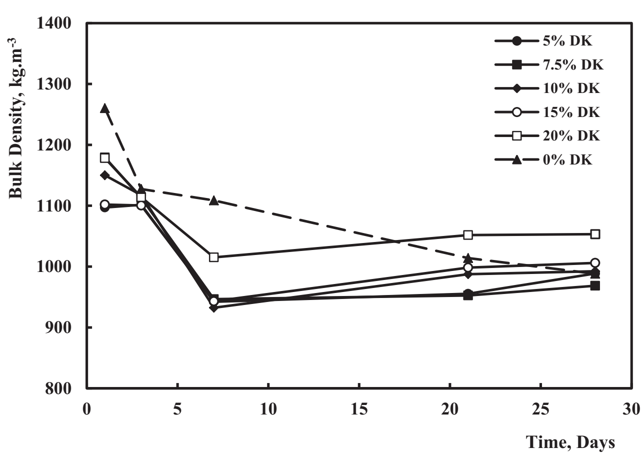



**Fig. 5.** Effect of substituting clay bricks dust binder by de-aluminated kaolin on bulk density (5% aluminum).

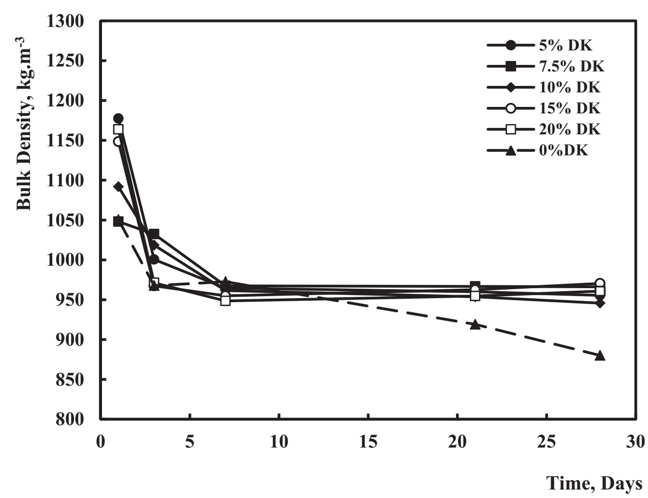



**Fig. 7.** Effect of substituting clay bricks dust binder by de-aluminated kaolin on bulk density (10% aluminum).

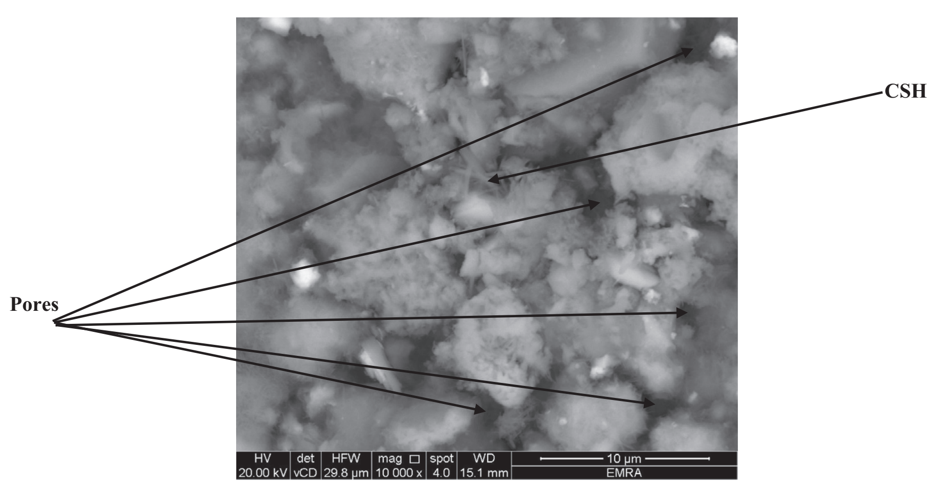



**Fig. 6.** Evidence of CSH needles formation and presence of pores (Specimen containing 5% Al, 10% DK cured for 7 days).

However, despite the increase in porosity due to hydrogen evolution, the formation of CSH needles contributed positively to the increase in strength, particularly at elevated DK levels. Fig. 8 illustrates the variation of strength with time for samples containing 5% aluminum. After an initial decrease in strength, due to hydrogen evolution, the formation of CSH needles caused the strength to increase reaching a value of about 1.4 MPa at elevated DK levels that shows little change after 7 days.

On increasing the amount of aluminum however, the formation of CSH needles could not compete with the increased porosity owing to more hydrogen formation. As a result, the strength values on using 10% aluminum were lower than if 5% aluminum were used (Fig. 9). The maximum strength obtained hardly exceeded 1 MPa. Further increase in aluminum level did not seriously affect this pattern, proving once more that adding up to 10% aluminum has exhausted all available calcium hydroxide so that any excess aluminum was left unreacted.

### 3.3. Effect of composition of geopolymers on the degree of polymerization

The compressive strength of geopolymers has often been regarded as a measure of the degree of geopolymerization [49].

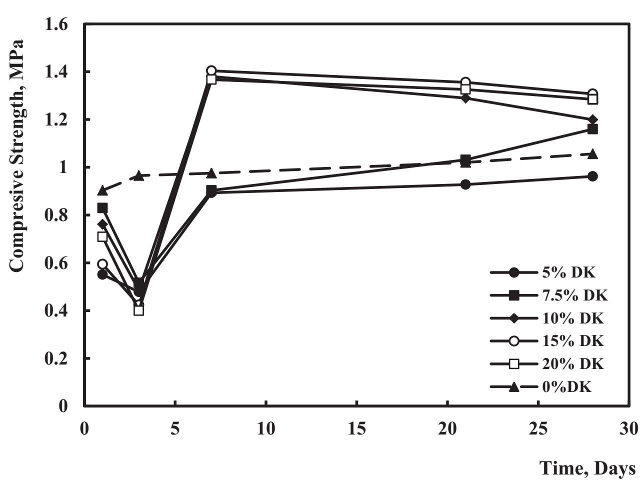



**Fig. 8.** Effect of substituting clay bricks dust binder by de-aluminated kaolin on compressive strength (5% aluminum).

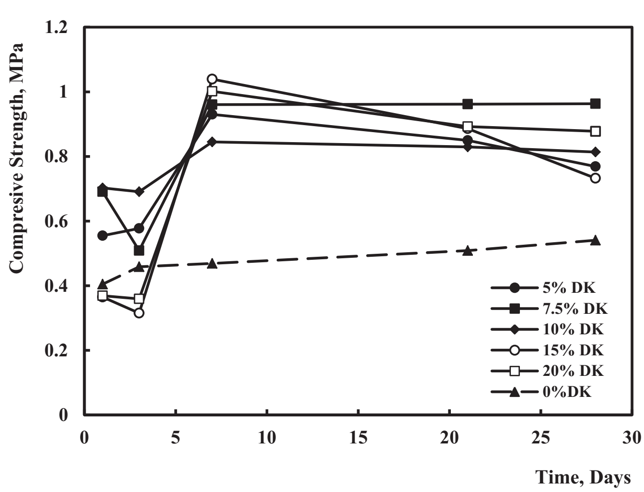



**Fig. 9.** Effect of substituting clay bricks dust binder by de-aluminated kaolin on compressive strength (10% aluminum).

In order to assess the effect of substituting part of the waste clay bricks dust by de-aluminated kaolin (DK), aluminum free samples were prepared with different levels of substitution. The degree of polymerization was determined, as previously indicated by the method proposed by Nikolic et al. [46]. Fig. 10 shows a slight improvement in the degree of geopolymerization after 28 days as more clay bricks dust was substituted by DK. When the 28 days specimens were tested for compressive strength, the obtained figures of strength were too close to be correlated with the degree of geopolymerization, their values ranging from 4.85 to 5.07 MPa.

### 3.4. Prediction of bulk density of the prepared geopolymer samples

An attempt was made to correlate the bulk density of produced samples to the percent alumina used, the curing time and the percent of substitution of clay bricks dust by DK. In this respect, the data of all prepared samples were inserted. The individual correlation coefficients shown in Table 2 seem to point out that the main parameters influencing the bulk density are the percent aluminum added and the curing time with little effect of substitution by DK.

A linear expression was deduced that correlated these variables that showed a multiple correlation coefficient = 0.73. It takes the form:

$$\rho_B = 1135.8 - 5.3 \times (\%\mathrm{Al}) - 3.7t - 0.243 \times (\%\mathrm{DK}) \tag{8}$$

Fig. 11 compares the calculated values to observed values of bulk density.

### 3.5. Thermal conductivity

The thermal conductivities of twelve specimens prepared with different proportions of aluminum and DK were determined after 28 days curing. Table 3 shows the values obtained. The table reveals that while the percentage DK did not seriously affect thermal conductivity, this was strongly influenced by the level of aluminum added. This was ascertained by the determination of the individual correlation coefficients between thermal conductivity and percent DK and aluminum (Table 4). On the other hand, a

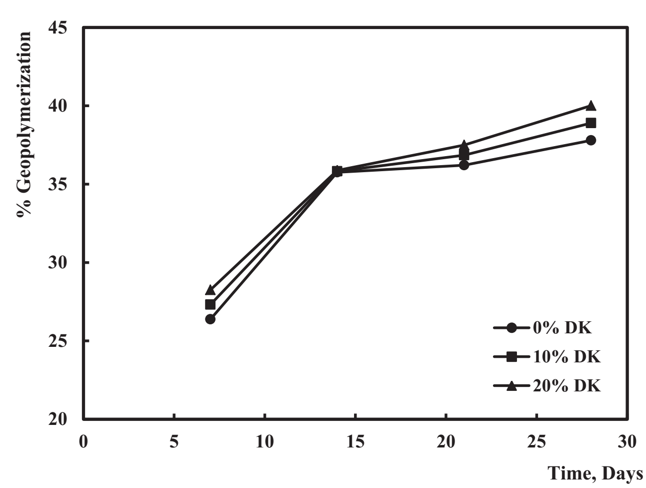



**Fig. 10.** Effect of substituting clay bricks dust binder by de-aluminated kaolin on the degree of geopolymerization.

**Table 2**
Correlation table of bulk density to different parameters.

| Parameter | % Aluminum | Time | % DK |
|---|---|---|---|
| Correlation Coeff. | −0.453 | −0.452 | −0.069 |

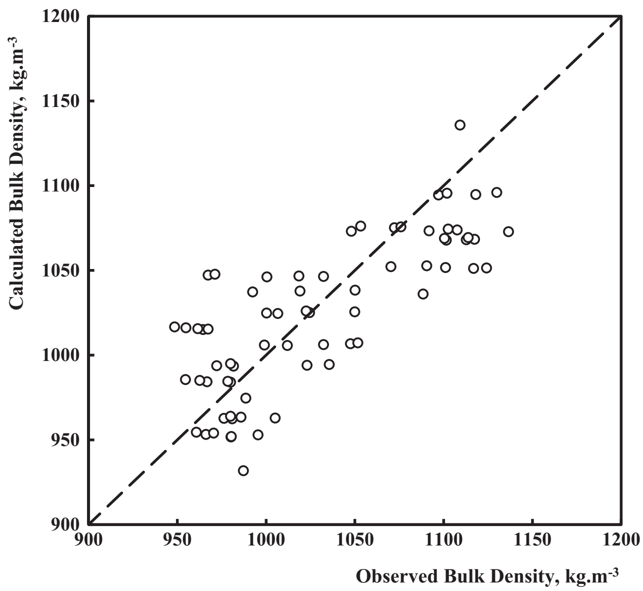



**Fig. 11.** Comparison between calculated and observed bulk densities.

**Table 3**
Thermal conductivity of geopolymer samples (W.K$^{-1}$.m$^{-1}$).

| DK% | 5 | 10 | 15 | 20 |
|---|---|---|---|---|
| 5% Al | 0.25 | 0.28 | 0.26 | 0.28 |
| 10% Al | 0.24 | 0.26 | 0.25 | 0.26 |
| 15% Al | 0.35 | 0.38 | 0.33 | 0.32 |

**Table 4**
Correlation table between thermal conductivity and % DK, % Al.

| Variable | %DK | % Al |
|---|---|---|
| *R* | 0.0395 | 0.6284 |

remarkable result could be observed, that is, the thermal conductivity for samples containing 15% aluminum are higher than those of samples with lower aluminum levels. This can be explained in light of the completion of pore formation by the aluminum – lime reaction by exhaustion of lime at about 10% aluminum addition. Any excess unreacted aluminum will definitely result in an increase in thermal conductivity, this metal being an excellent heat conductor. A SEM micrograph coupled with the corresponding EDX pattern was performed on a specimen containing 15% aluminum. It showed the presence of unreacted metal as evidenced by the glossy particles appearing in the SEM micrograph and EDX results shown in Fig. 12.

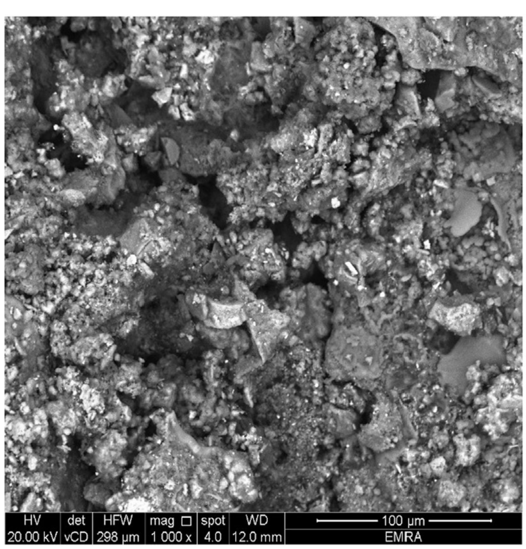
On the other hand, samples containing 5 or 10% aluminum exhibited lower thermal conductivities reaching values as low as 0.24 W.m$^{-1}$.K$^{-1}$. These values of thermal conductivity are lower than those of conventional perforated clay bricks (About 0.8 W.m$^{-1}$.K$^{-1}$) and even clay bricks with induced porosity (about 0.4 W.m$^{-1}$.K$^{-1}$) [50,51].

## 4. Conclusion

Optimally, a geopolymer brick to be used for insulation purposes should possess a low bulk density, a low thermal conductivity while keeping a reasonable mechanical strength.

Accordingly, following the experimental results obtained, substituting clay bricks waste dust by de-aluminated kaolin by 15% and using 5% aluminum seems to be a good choice. The prepared brick samples exhibit a bulk density of about 1000 kg.m$^{-3}$, a 28 days compressive strength of 1.4 MPa and a thermal conductivity of 0.26 W.m$^{-1}$.K$^{-1}$. These results obtained are in agreement with those obtained by Silva et al. [52], who prepared foamed concrete samples of porosity ≈50%, of bulk density 940 kg.m$^{-3}$ and a thermal conductivity of 0.24 W.m$^{-1}$.K$^{-1}$ although these authors obtained a higher value of compressive strength for that mix (>4 MPa), presumably because of the incorporation of Portland cement and silica fume into the mix.

The economic merit of the prepared geopolymer bricks is that they are mainly constituted of costless waste materials, namely, fine clay bricks waste, de-aluminated kaolin waste, aluminum trimmings and spent slaked lime waste from the acetylene industry.



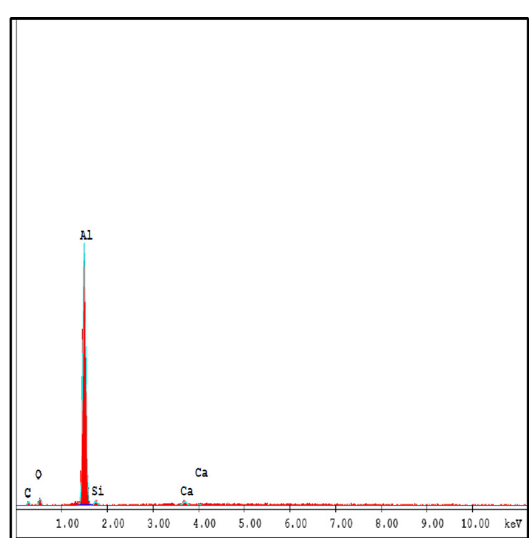




**Fig. 12.** Evidence of the presence of unreacted aluminum metal.

## Declaration of Competing Interest

None declared.

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